

Fraud Dispute & Chargebacks

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Problems

Chargeback disputes currently take weeks to resolve because evidence from the cardholder and the merchant is evaluated manually, inconsistently, and without transparency to either party.

Cardholders who have genuinely experienced non-delivery, unauthorized charges, or improper service often face slow, opaque investigations before receiving a fair refund.

Merchants face financial loss — not only from the disputed amount, but from associated penalty fees — when they are unable to present their evidence effectively, including against invalid or fraudulent claims.

There is currently no fast, fair, and transparent way to weigh both parties' evidence and reach a resolution — leaving both cardholders and merchants dissatisfied with the process.

Proposed Solution

An AI-powered Dispute Resolution Module that enables cardholders and merchants to securely submit evidence for an existing chargeback case. The system autonomously collects, structures, and compares evidence from both parties, requests additional information when required, and generates a transparent decision recommendation with reasoning. For complex or low-confidence cases, the dispute is escalated to an issuer analyst for final review.

Expected Social Impact

For Cardholders

- Faster dispute resolution (days/weeks → minutes/hours)
- Transparent reasoning behind every decision

- Reduced effort in repeatedly submitting documents

For Merchants

- Faster response to disputes through structured evidence requests
- Reduced revenue loss due to invalid/friendly fraud chargebacks
- Standardized evidence submission process

For Issuer

- Reduced manual investigation workload
- Consistent decision-making across analysts

Business Impact

- Reduced operational cost of manual investigations
- Lower chargeback processing time
- Reduction in friendly fraud losses
- Increased merchant trust in the dispute process
- Improved customer retention through transparent decisions
- Higher analyst productivity (one analyst can handle more cases)

Success Metrics

Operational Metrics

- Average dispute resolution time
- Average evidence collection time
- Number of cases handled per analyst
- Percentage of disputes resolved without human intervention

AI Metrics

- Evidence extraction accuracy
- Decision recommendation accuracy

- False positive / false negative rate
- AI confidence calibration

User Metrics

- Customer Satisfaction (CSAT)
- Merchant Satisfaction
- Average response time from customer/merchant
- Number of repeated evidence requests

Business Metrics

- Reduction in operational investigation cost
 - Reduction in friendly fraud losses
 - Chargeback reversal rate
 - Percentage reduction in unresolved disputes
-

Implementation Approach

System Overview

The proposed solution is an **AI-powered dispute resolution module** that automates the chargeback investigation process between the **cardholder** and the **merchant**. Traditionally, disputed transactions are manually reviewed by issuer analysts who collect evidence from both parties, verify compliance with card-network policies, and determine whether a chargeback should be accepted or contested. This manual workflow is time-consuming and requires significant human effort.

The proposed solution transforms this process into an **agentic AI workflow** that autonomously collects, structures, analyzes, and reasons over dispute evidence. The system is capable of resolving straightforward disputes with minimal human intervention while escalating complex or low-confidence cases to a human issuer analyst.

User Workflow

The system consists of three stakeholders: **Cardholder**, **Merchant**, and optionally an **Issuer Analyst** for human-in-the-loop review.

Cardholder Workflow

- The cardholder selects a disputed transaction and submits a dispute request through a structured form.
- The complaint is automatically classified into a standardized **chargeback reason code** (e.g., Item Not Received, Unauthorized Transaction, Item Not as Described).
- The reason code determines the required evidence checklist and investigation workflow.

Merchant Workflow

- Once the dispute is registered and validated, the merchant is notified.
- The merchant is requested to submit evidence relevant to the identified dispute category.
- Both parties can track the investigation status and receive requests for additional information if required.

Evidence Collection

After the dispute is created, the platform begins collecting evidence from multiple sources.

Cardholder Evidence

- Complaint description
- Images
- Receipts
- Screenshots
- Communication history
- Additional supporting documents

Merchant Evidence

- Order records

- Payment confirmation
- Invoice
- Shipping and tracking history
- Delivery confirmation
- Customer communication
- Merchant policies

Issuer Information

To reduce manual effort, the system also retrieves existing transaction metadata maintained by the issuer, including transaction details, timestamps, merchant information, authorization records, and payment status.

The evidence required from each party is dynamically determined based on the identified **reason code**.

AI Processing

Once evidence has been collected, the AI pipeline begins processing it.

The system extracts structured information from both structured and unstructured documents using document understanding techniques. Rather than treating PDFs, images, or transaction statements as isolated files, the extracted information is normalized into a common representation consisting of:

- Entities
- Facts
- Events
- Dates and timelines
- Relationships between evidence

This structured representation serves as the foundation for downstream reasoning.

Knowledge Graph & Reasoning

Instead of directly asking a Large Language Model to decide the dispute outcome, the extracted information is transformed into a **knowledge graph**.

The graph represents interconnected information including:

- **Entities:** Cardholder, Merchant, Order, Payment, Shipment, Issuer
- **Events:** Payment, Shipment, Delivery, Refund Attempts, Dispute Filing
- **Evidence:** Invoice, Delivery Proof, Tracking History, Communication Logs
- **Policies:** Chargeback reason-code requirements, issuer policies, evidence sufficiency rules

Using this knowledge graph, multiple specialized AI agents collaboratively evaluate the dispute.

Intake Agent

Classifies the dispute into the appropriate standardized reason code.

Evidence Agent

Determines the required supporting documents and evaluates evidence completeness.

Policy Agent

Verifies compliance with issuer policies and chargeback requirements.

Timeline Agent

Analyzes the chronological sequence of events and validates submission deadlines.

Contradiction Agent

Identifies inconsistencies or conflicts between customer and merchant claims.

Decision Agent

Aggregates insights from all reasoning agents and generates a confidence-backed recommendation.

Decision & Explainability

The final recommendation includes:

- Recommended dispute outcome
- Confidence level

- Evidence comparison
- Missing or contradictory evidence
- Policy references used during reasoning
- Explainable decision report describing why the recommendation was generated

This transparency enables both stakeholders and issuer analysts to understand the rationale behind every decision.

Human Escalation (Optional)

For disputes involving:

- Low confidence recommendations
- Conflicting evidence
- Insufficient documentation
- High-value transactions

the case is automatically escalated to an **issuer analyst**.

Technical Details

1. Tech Stack

Layer	Technology
Frontend	React.js, Tailwind CSS
Backend API	FastAPI
Authentication	JWT
AI/NLP	spaCy, Hugging Face Transformers, OCR
LLM	Groq API / Hugging Face Inference
Agent Framework	LangGraph
Database	PostgreSQL
File Storage	ImageKit (or AWS S3)
Cloud Deployment	AWS

2. Assumptions

Data Assumptions

- Cardholder, merchant, and transaction details are assumed to be available through the payment ecosystem.
- Both parties are able to upload valid supporting documents during the dispute process.

AI Assumptions

- OCR and NLP models can reliably extract structured information from common document formats (PDFs, invoices, screenshots, images).
- The AI system provides recommendation-based decisions with explainable reasoning.
- Highly complex disputes may require human review.

System Assumptions

- The dispute is raised within the permitted chargeback time window.
- Standardized chargeback reason codes and issuer policies are available in structured format.

Project Scope

- Only completed transactions are considered.
- Real-time fraud prevention before payment authorization is outside the project scope.
- The project focuses only on post-payment dispute resolution.

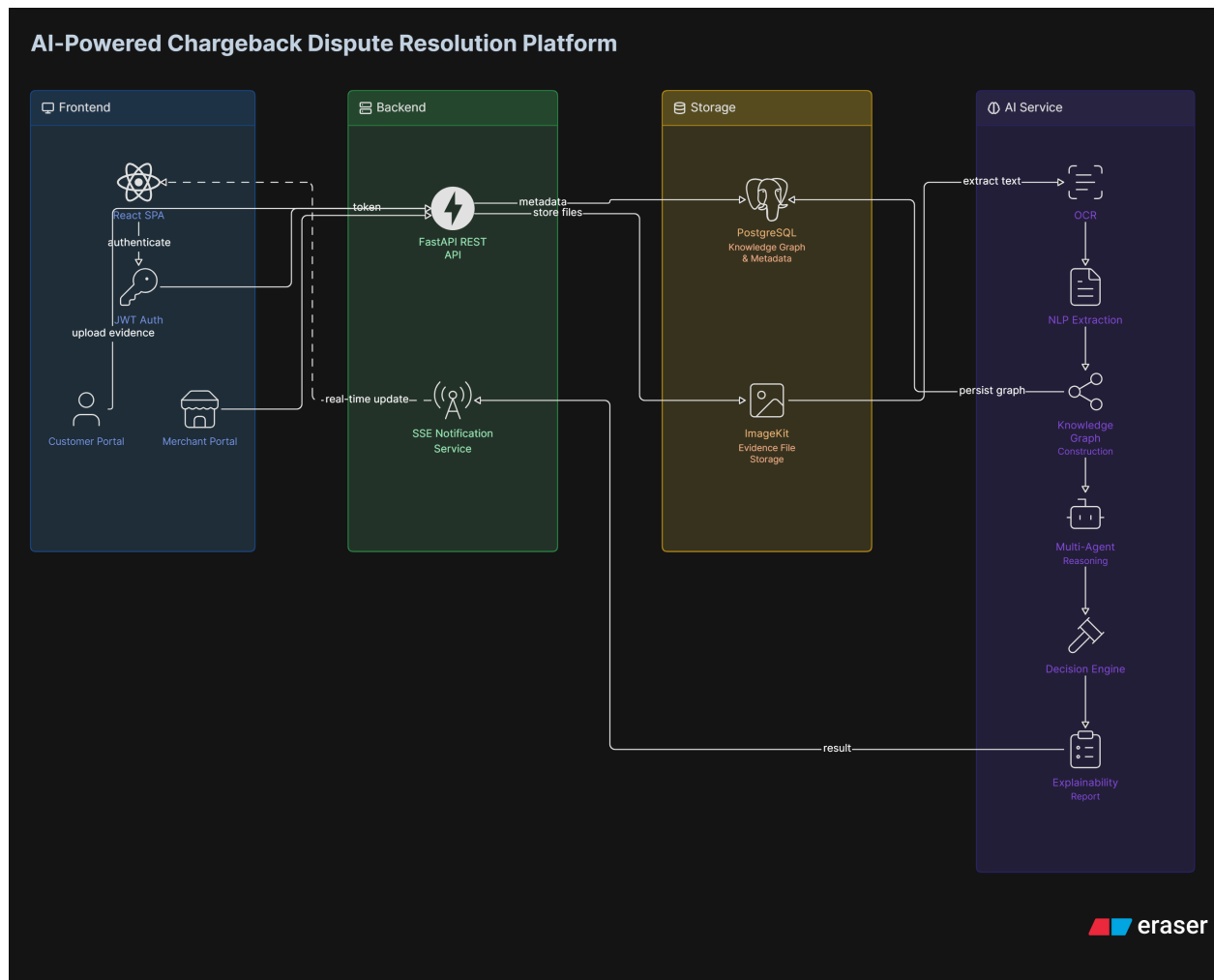
3. Scalability

- The MVP is designed to handle a limited number of concurrent dispute cases.
- PostgreSQL is used to represent the knowledge graph using relational tables. For significantly larger graphs, a dedicated graph database (e.g., Neo4j) can be adopted.
- AI inference currently relies on hosted APIs and free-tier resources, which may limit throughput under heavy workloads.

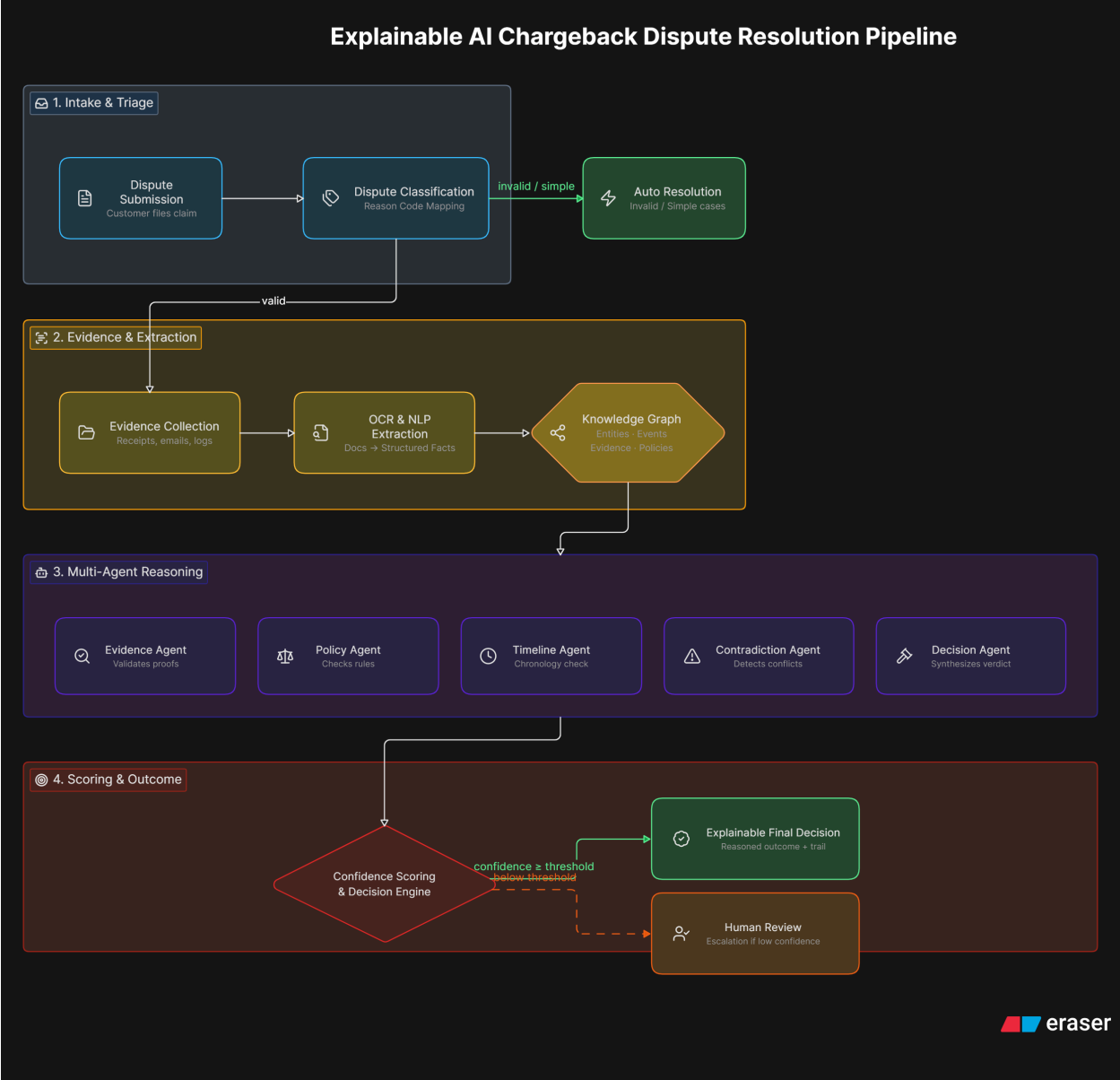
- The architecture is modular, allowing OCR, reasoning agents, or storage services to be replaced independently in future versions.
- Additional evidence types (emails, videos, audio, chat logs) can be incorporated by extending the extraction pipeline.

4. Architecture

System Architecture

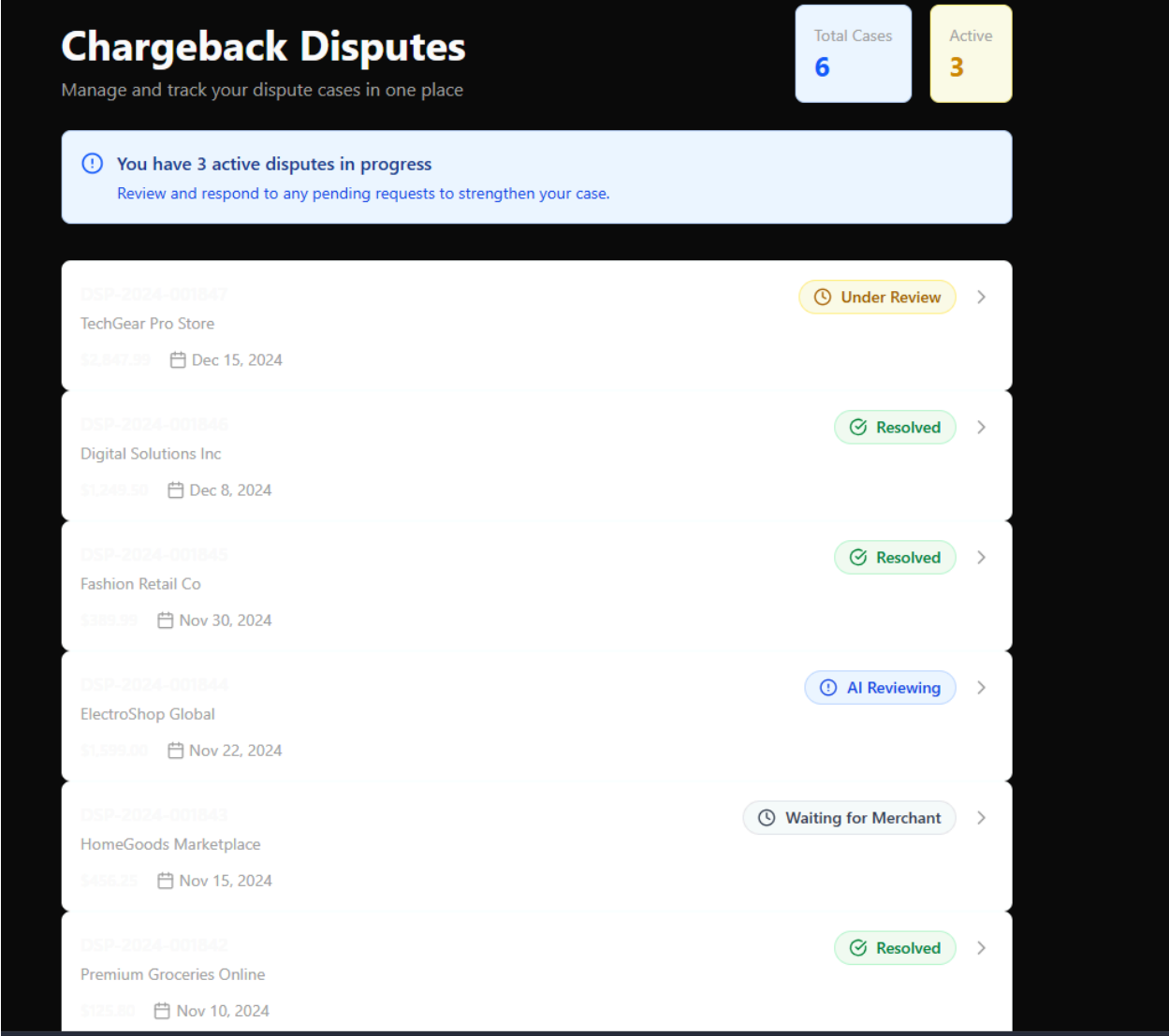


AI Pipeline



Prototype Screenshots

Customer Dashboard



Customer Dashboard/#case1

Merchant Dashboard



Chargeback Portal

Manage and respond to chargeback disputes efficiently

Total Cases

9



Active

3



Waiting for You

3



Resolved

3



All Cases ▾

Case ID

CHG-2024-001

Waiting for You

Customer

Sarah Johnson

Amount

\$249.99

Date

Jan 15, 2024

Reason

Unauthorized Transaction

2 files submitted

→

Case ID

CHG-2024-002

Under Review

Customer

Michael Chen

Amount

\$85.50

Date

Jan 10, 2024

Reason

Item Not Received

3 files submitted

→

Case ID

CHG-2024-003

Resolved

Customer

Emily Rodriguez

Amount

\$125.00

Date

Jan 12, 2024

Reason

Service Not Provided

3 files submitted

→

Case ID

CHG-2024-004

AI Reviewing

Customer

James Wilson

Amount

\$599.99

Date

Jan 8, 2024

Reason

Duplicate Charge

2 files submitted

→

Merchant Dashboard/#case1

[← Back to Cases](#)

Case ID

CHG-2024-003

Resolved

CUSTOMER

Emily Rodriguez

AMOUNT

\$125.00

DISPUTE REASON

Service Not Provided

TRANSACTION DATE

Jan 12, 2024

Customer's Claim

DISPUTE REASON

Service Not Provided

CUSTOMER'S EXPLANATION

Service quality did not meet expectations. The product arrived damaged.

Recent Updates

✓ Case resolved in favor of merchant. Refund processed.

Jan 14

Submitted Evidence

File Name	Type	Size	Uploaded	Action
Quality_Photos.zip	Archive	2 MB	2024-01-14	View
Customer_Messages.pdf	Pdf	332.03 KB	2024-01-14	View
Refund_Agreement.pdf	Pdf	112.3 KB	2024-01-14	View

Timeline

- Order delivered**
Jan 12, 2024, 06:50 PM
Product received by customer
- Complaint received**
Jan 13, 2024, 03:35 PM
Quality issue reported
- Case resolved**
Jan 14, 2024, 09:15 PM
Refund issued to customer



Case Resolution